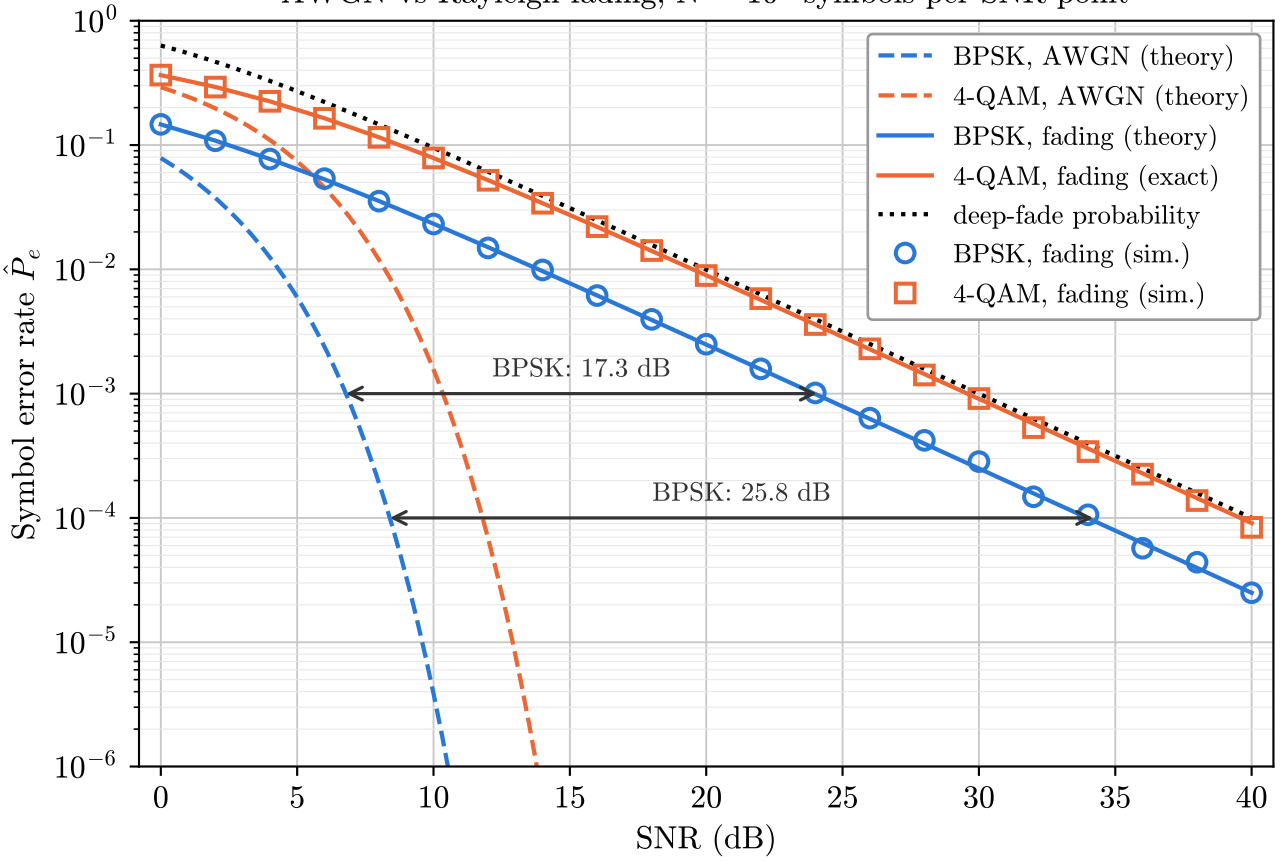
AWGN vs Rayleigh fading, N = 10^6 symbols per SNR point
Symbol error rate $\hat{P}_e$
SNR (dB)
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
0
5
10
15
20
25
30
35
40
BPSK, AWGN (theory)
4-QAM, AWGN (theory)
BPSK, fading (theory)
4-QAM, fading (exact)
deep-fade probability
BPSK, fading (sim.)
4-QAM, fading (sim.)
BPSK: 17.3 dB
BPSK: 25.8 dB